

Generation Interconnection Request System Impact Study Report

For Queue Project Q17-0874

Palisades 230 kV
July 18, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0874. The Interconnection Customer proposes a 351.3 MW solar photovoltaic facility interconnecting at the Palisades 230 kV substation in the Michigan area. The request was received on October 08, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 290 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,953,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The BARRETT – HEBO SUBSTATION 230kV line is overloaded to 109.1% of its 560 MVA emergency rating for the loss of the WINTHROP – HEBO 230kV. The project contributes 5.8% DFAX (20.5 MW) to this overload.

TREGO – BEAVER 500kV Line: 103.5% loading (4,242 MVA) following the outage of the ADAMS – ONANCOCK DIESEL GENERATION FACILITY 230kV; DFAX = 30.7%.

The ADAMS – ONANCOCK DIESEL GENERATION FACILITY 230kV is overloaded to 102.9% of its 2,118 MVA emergency rating for the loss of the TREGO – BEAVER 500kV Line. The project contributes 15.8% DFAX (55.3 MW) to this overload.

For the loss of TREGO – BEAVER 500kV, the NEW DEPARTURE – MASON ST SUBSTATION 230kV loads to 101.9% of its 1,109 MVA rating. Project DFAX: 7.7%.

For the loss of NEW DEPARTURE – MASON ST SUBSTATION 230kV L, the HERALD-TWIN CITIES – MASON ST 230kV loads to 101.9% of its 759 MVA rating. Project DFAX: 5.3%.

ADAMS – PICKFORD 230kV: 101.9% loading (1,089 MVA) following the outage of the BEAVER – TREGO 500kV; DFAX = 8.0%.

HERALD-TWIN CITIES – PICKFORD 230kV: 101.5% loading (1,423 MVA) following the outage of the BEAVER – TREGO 500kV; DFAX = 9.6%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BARRETT – HEBO SUBSTATION 230kV line	Reconductor line	\$5,984,000	3% (\$160,000)
BEAVER – TREGO 500kV L	Reconductor line	\$143,780,000	3% (\$3,938,000)
ADAMS – ONANCOCK DIESEL GENERATION FACILITY 230kV	Reconductor line	\$22,669,000	3% (\$624,000)
NEW DEPRTURE – MASON ST SUBSTATION 230kV	Reconductor line	\$12,944,000	3% (\$357,000)
HERALD-TWIN CITIES – MASON ST 230kV	Reconductor line	\$12,547,000	3% (\$408,000)
ADAMS – PICKFORD 230kV	Reconductor line	\$5,704,000	3% (\$157,000)
HERALD-TWIN CITIES – PICKFORD 230kV	Reconductor line	\$11,234,000	3% (\$309,000)
	Total allocated to project		\$5,953,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.